

Regularized Multivariate Two-way Functional Principal Component Analysis

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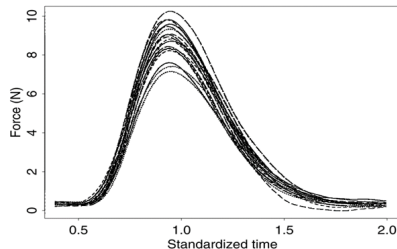
Outline

- 1 Introduction & Background
- 2 A SVD Approach for Regularized Multivariate FPCA
- 3 Two-way Regularized Multivariate FPCA
- 4 Conclusion & Future Work

Introduction

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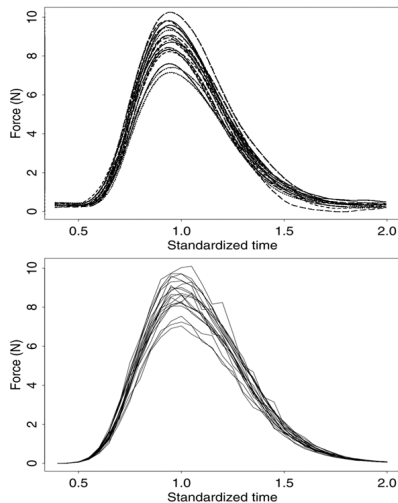
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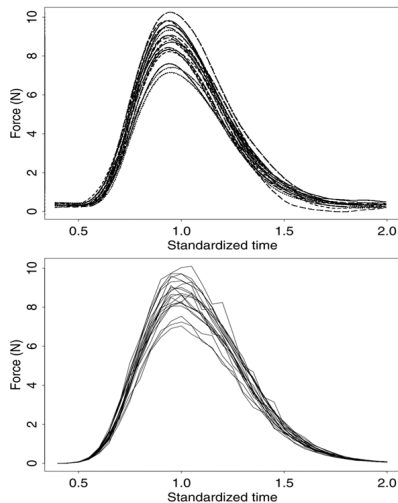
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- These data are often recorded at discrete time points or grid locations, even though they originate from continuous processes in areas like engineering, finance, environmental science, and healthcare.
- Functional Data Analysis (FDA) is a statistical framework that treats these observations as realizations of smooth underlying functions, allowing for more accurate modeling and interpretation of continuous processes.



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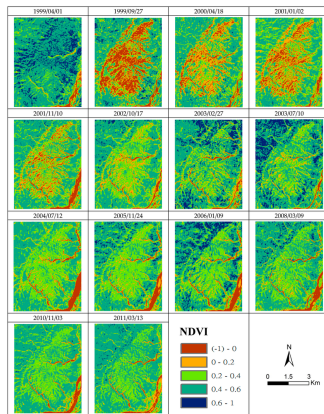
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- Impact: More adaptable, robust, and applicable across diverse scientific and business problems.

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 - Effective for dimension reduction in complex datasets.

Mathematical Foundation of FPCA

- **Goal:** Identify functional directions that maximize variance (low-rank approximation of functional data). For functional data $X \in \mathbb{R}^{n \times m}$ contains the discretized functional observations (rows correspond to subjects, columns to grid points), $v \in \mathbb{R}^m$ represents the estimated principal component (function), and $u \in \mathbb{R}^n$ denotes the associated principal component scores.

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- **Optimization steps:**

$$\text{Fix } v : u = \frac{Xv}{v^\top v} \quad \text{and} \quad \text{Fix } u : v = \frac{X^\top u}{u^\top u}$$

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- Algorithms: Based on iterative **power method** and **thresholding** updates.

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- Consider the SVD of X as $X = UDV^\top$, where U and V have orthonormal columns and D is diagonal with ordered singular values. In particular, for $X = u d v^\top$, v is the first principal component and $u = u d$ gives the associated scores. With these representations, the power algorithm (described below) converges quickly, typically in only a few iterations.

Power Algorithm

Algorithm: Penalized Power Iteration

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- For notational convenience, we define $S(\alpha) = (I + \alpha\Omega)^{-1} \in \mathbb{R}^{m \times m}$, which simplifies expressions involving regularization. The penalty matrix Ω is set up so that larger values of the quadratic form $v^\top \Omega v$ mean rougher functions. This means that it penalizes functions that change quickly between time points.

Tuning Smoothness Parameters

- To select the optimal tuning parameters α efficiently, one can use a traditional Cross-Validation (CV) criterion and a computationally efficient closed-form Generalized Cross-Validation (GCV) criterion:

$$\text{CV}(\alpha) = \frac{1}{m} \sum_{j=1}^m \frac{\left[\{(I - S(\alpha))(X^T u)\}_{jj} \right]^2}{(1 - \{S(\alpha)\}_{jj})^2},$$

where $\{\cdot\}_{jj}$ denotes the j -th diagonal element.

$$\text{GCV}(\alpha) = \frac{1}{m} \frac{\|(I - S(\alpha))(X^T u)\|^2}{\left(1 - \frac{1}{m} \text{tr}\{S(\alpha)\}\right)^2}.$$

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- Sparse FPCA formulation [Shen and Huang, 2008]:

$$\min_{u,v} \|X - uv^T\|_F^2 + p_\gamma(v)$$

where $p_\gamma(v)$ is a sparsity-inducing penalty.

Sparsity penalties

- **Lasso (Soft-thresholding):**

$$p_{\gamma}^{\text{soft}}(|\theta|) = 2\gamma|\theta|, \xrightarrow{\text{minimizer}} h_{\gamma}^{\text{soft}}(y) = \text{sign}(y)(|y| - \gamma)_+$$

- **Hard thresholding:**

$$p_{\gamma}^{\text{hard}}(|\theta|) = \gamma^2 I(|\theta| \neq 0), \xrightarrow{\text{minimizer}} h_{\gamma}^{\text{hard}}(y) = I(|y| > \gamma) y$$

- **SCAD penalty:**

$$p_{\gamma}^{\text{SCAD}}(|\theta|) = \begin{cases} 2\gamma|\theta|, & |\theta| \leq \gamma, \\ \frac{\theta^2 - 2a\gamma|\theta| + \gamma^2}{a-1}, & \gamma < |\theta| \leq a\gamma, \\ \frac{(a+1)\gamma^2}{2}, & |\theta| > a\gamma, \end{cases} \xrightarrow{\text{minimizer}} h_{\gamma}^{\text{SCAD}}(y) = \begin{cases} \text{sign}(y)(|y| - \gamma)_+, & |y| \leq 2\gamma, \\ \frac{(a-1)y - \text{sign}(y)a\gamma}{a-2}, & 2\gamma < |y| \leq a\gamma, \\ y, & |y| > a\gamma, \end{cases}$$

where $a = 3.7$ (Fan and Li [2001]).

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 - ➌ **Normalize Right Singular Vector:** $v \leftarrow \frac{v}{\|v\|}$

Cross-Validation for Sparsity Selection

- **Sparsity parameter:** Tuning parameter controlling number of non-zero loadings in v ($0 =$ dense, $p =$ full sparsity).

Algorithm: K-fold CV Tuning Parameter Selection - Degree of sparsity

- 1 Randomly group the rows of side-by-side data matrix X into K roughly equal-sized groups, denoted as $X^1, \dots, X^K$.

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 - 2 Calculate the K-fold CV scores defined as: (N is the number of grid points in X^k)

$$CV_j = \sum_{k=1}^K \frac{\|X^k - u^{-k}(j)v^k(j)\|^2}{N}$$

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- 3 Select the degree of sparsity as $j_0 = \arg \min \{CV(j)\}$.

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- More stable & applicable to high-dimensional multivariate FDA

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- **Block structure example:**

$$X_i = \begin{bmatrix} x_{1i}(t_{i1}) & \cdots & x_{1i}(t_{i,m_i}) \\ \vdots & \ddots & \vdots \\ x_{ni}(t_{i1}) & \cdots & x_{ni}(t_{i,m_i}) \end{bmatrix}, \quad \mathbf{X} = [X_1 \ X_2 \ \cdots \ X_p]$$

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- The penalized reconstruction error is

$$\min_{u,v} \|\mathbf{X} - uv^\top\|_F^2 + \alpha^\top (v^\top \mathbf{\Omega} v),$$

where $\alpha = (\alpha_1, \dots, \alpha_p)^\top$ controls smoothness.

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 - The smoothing parameter α is selected via generalized cross-validation (GCV), defined as

$$\text{GCV}(\alpha) = \frac{1}{M} \frac{\|(I - \mathbf{S}(\alpha))(\mathbf{X}^\top u)\|^2}{\left(1 - \frac{1}{M}\text{tr}\{\mathbf{S}(\alpha)\}\right)^2}. \quad (1)$$

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- Algorithm: CV Tuning for Sparsity and equation (1) are used to tune the sparsity level via K-fold cross-validation (CV) and the smoothing parameter via generalized cross-validation (GCV), respectively.

Two-way Regularized MFPCA

Conclusion & Future Work

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